

Multi-Axis Failure Modes in Simulator Transfer

Diagnostics from Discrete Maneuver Tokens and Grounded Selection

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Waymo Open Dataset E2E Challenge / CoRL 2027 branch

The Core Claim

Internal closed-loop success is not a reliable transfer proxy.

We find that simulator transfer failure decomposes into separate axes:

- geometry: offroad and route reconstruction
- safety: collision and proximity violation
- progress: distance and route completion
- lane adherence: wrong-lane or lane-violation behavior

No evaluated intervention is co-monotone across all axes in AlpaSim.

Why This Matters Now

Recent VLA work moves toward physically grounded latent reasoning.

LaST-VLA argues that autonomous-driving VLA reasoning should use 3D geometric priors

and world-model dynamics, not only text chain-of-thought.

It reports NAVSIM v1/v2 gains through latent spatio-temporal reasoning.

Our question is narrower and diagnostic:

If a policy or selector is internally strong, which transfer axes actually survive when the observation and dynamics interface changes?

Reference: [LaST-VLA, arXiv:2603.01928](#)

Experimental Stack

Layer	Purpose	Evidence
Custom closed-loop simulator	Controlled long-tail AV stress lab	LHS generator, DAgger, proxy perturbations
AlpaSim adapter	External sim-to-sim validation on WOD-E2E clips	10 matched clips, partial 13-scene refresh
WOD candidate-selection track	Real-data grounding probe	479 validation frames, segment-grouped CV

The tracks are deliberately separated.

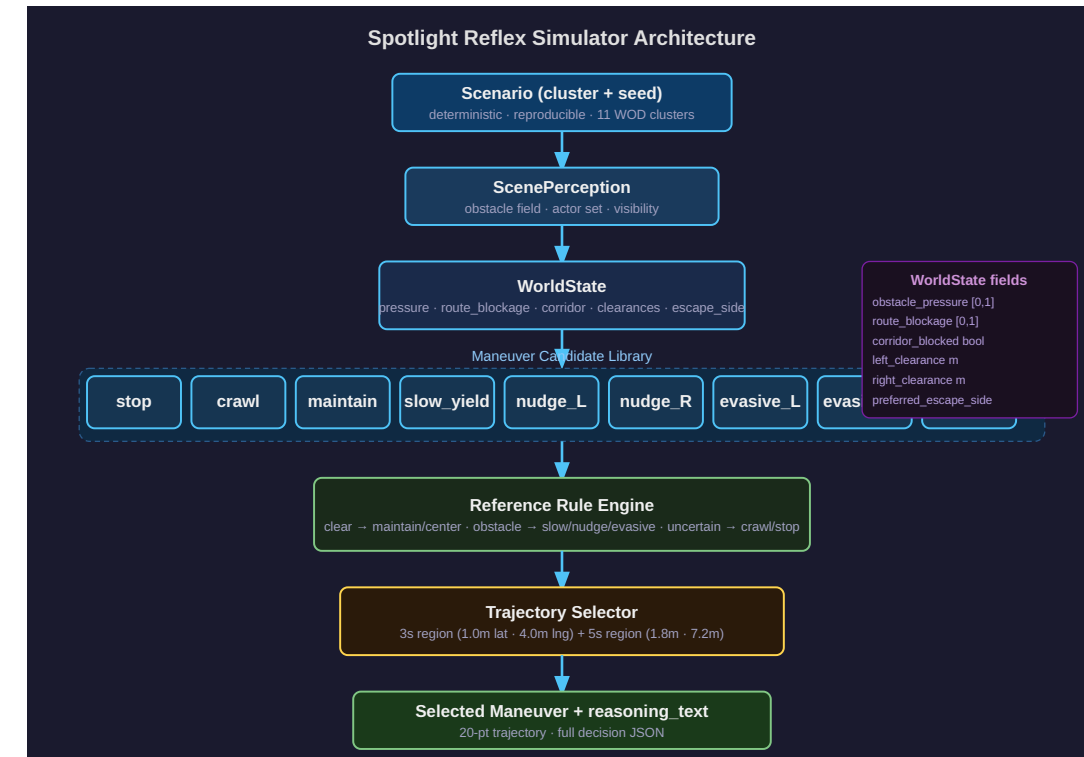
Simulator metrics are not used to tune WOD selector results.

What the Simulator Tests

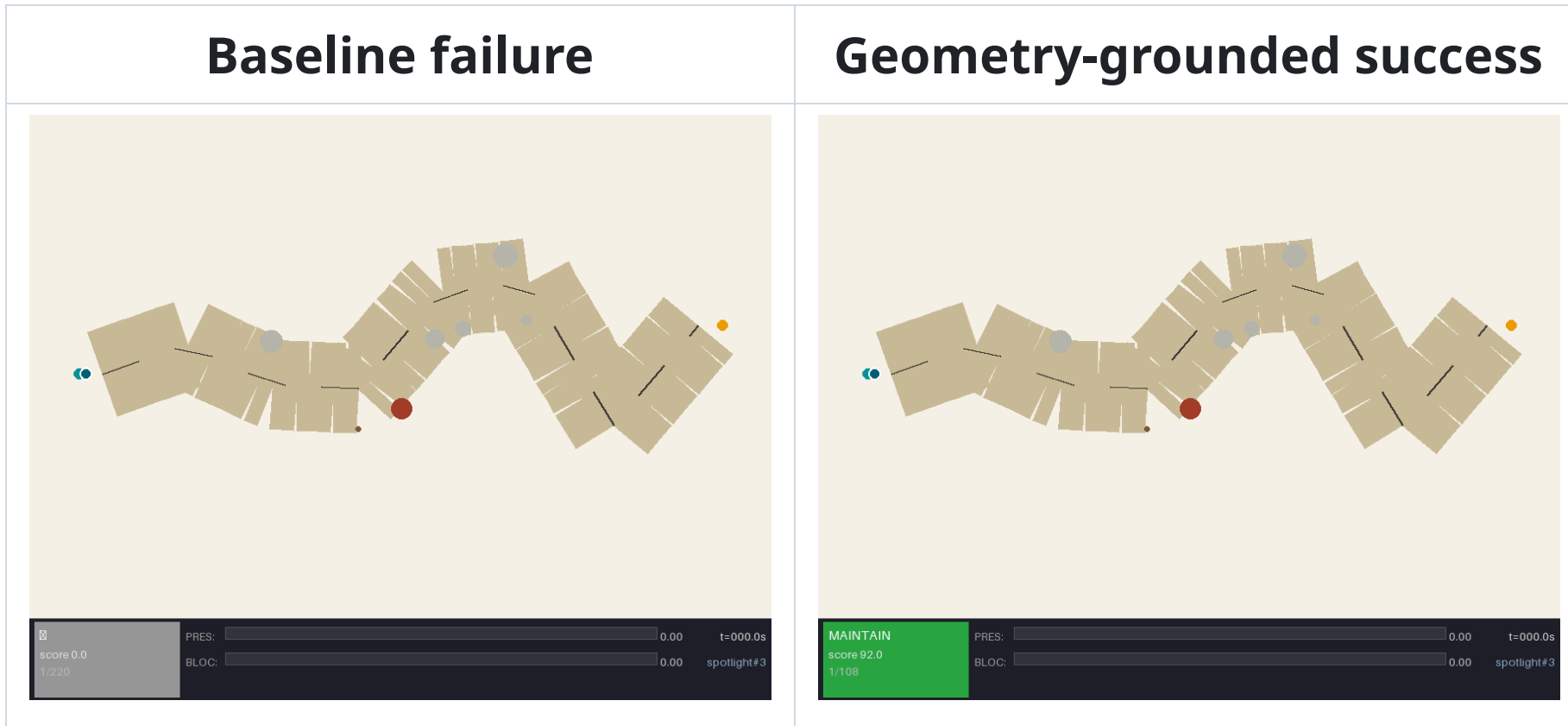
The internal simulator is a controlled micro-lab, not a realism claim.

It lets us isolate:

- discrete token action spaces
- rule-based geometric selection
- BC and DAgger imitation variants
- proxy-state corruption with true-state evaluation
- matched seeds across policies and perturbations



Visual Failure and Success Cases



Same scenario family: wrong-way actor under low visibility.

The baseline extrapolates into the hazard. Spotlight selects an evasive token from

Policy Interface

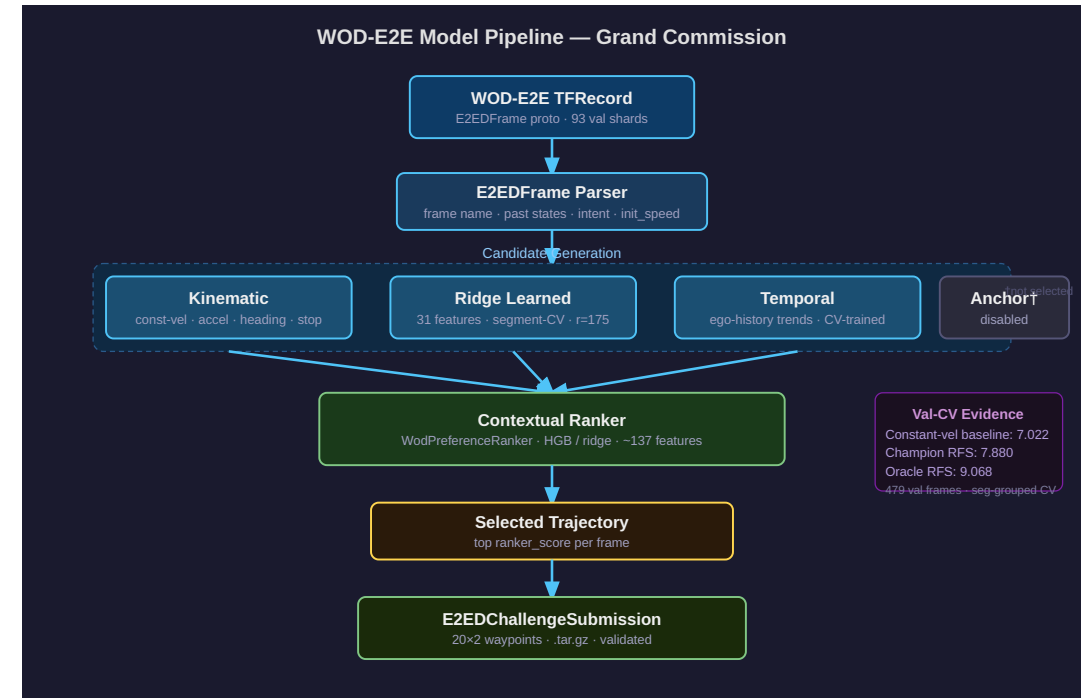
State:

- 10 scalar geometric features for learned policies
- six core world-state scalars for Spotlight

Action:

- 9 ManeuverTokens
- stop, crawl, maintain, slow-yield
- nudge left/right, evasive left/right, lane-recover

The policy selects a token; the



Internal Result: DAgger Closes the Local Frontier

Held-out Latin-hypercube sweep: 12 profiles, seeds 1-10, Gauntlet / Adversarial / Hidden.

Each agent: 1080 closed-loop rollouts.

Agent	Overall pass / PV	Gauntlet	Adversarial	Hidden
Continuous-BC	77.69 / 3.52	77.50 / 4.31	75.42 / 2.08	83.33 / 1.67
Token-BC	76.85 / 4.26	77.64 / 4.86	71.67 / 2.92	82.50 / 3.33
		79.17 /		81.67 /

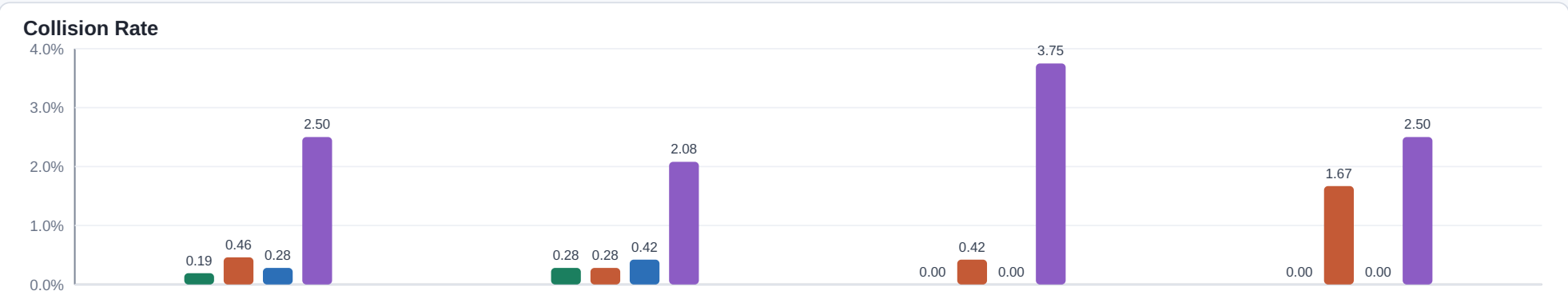
Internal Result: More DAgger Is Not Monotonic

Policy	Overall pass / PV	Gauntlet	Adversarial	Hidden
Token-DAgger, 1-step	93.98 / 0.46	96.81 / 0.56	87.50 / 0.42	90.00 / 0.00
Token-DAgger, 2-step	94.54 / 0.19	96.25 / 0.28	90.83 / 0.00	91.67 / 0.00
Token-DAgger, 3-step	92.50 / 0.46	94.31 / 0.28	87.92 / 0.42	90.83 / 1.67
3-step source decay	93.06 / 0.28	94.86 / 0.42	89.17 / 0.00	90.00 / 0.00
3-step no inverse	89.35 / 2.50	92.08 /	82.50 / 3.75	86.67 /

Dagger Aggregation Visual

Dagger Aggregation Ablations on Held-Out Latin-Hypercube Sweep

Two-step Dagger remains the frontier. Source decay partially recovers the three-step collapse; removing inverse-frequency weights breaks closed-loop safety.



Negative Result: Scoring Ego Trajectories Was Not Enough

We trained trajectory-informed scorers over candidate futures with:

- curvature and heading deltas
- path length, acceleration, jerk proxies
- minimum dynamic clearance and time-to-closest-approach
- signed lateral and longitudinal miss distances

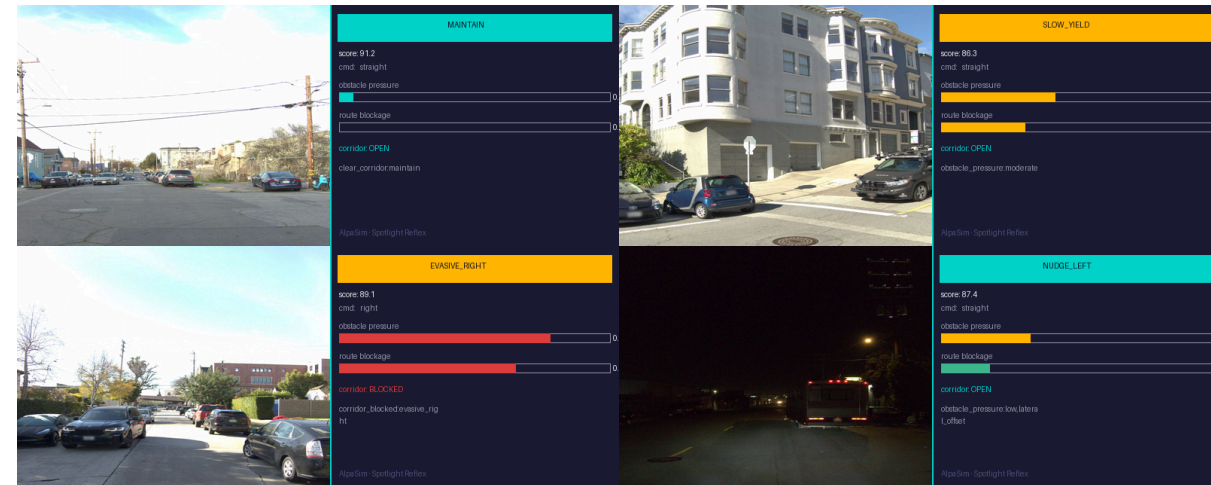
Policy	Gauntlet pass / PV	Adversarial	Hidden
Interaction scorer, learner states	86.81 / 1.94	85.00 / 1.67	89.17 / 1.67
Interaction scorer, learner states + ...	86.81 / 1.94	85.00 / 1.67	89.17 / 1.67

External Diagnostic: AlpaSim

AlpaSim uses real WOD-E2E front-camera frames and an adapter that reconstructs:

- brightness
- route command
- ego dynamics
- sparse hazard primitives
- geometric scalars consumed by the policy

This is a transfer diagnostic, not a



AlpaSim Bridge

The external bridge keeps the policy binary fixed.

What changes is the interface:

- camera and route observations become proxy geometric scalars
- sparse hazards become obstacle and actor primitives
- token selection is evaluated under AlpaSim execution

This is exactly where sim-to-sim mismatch is exposed.

AlpaSim Matrix: One Fix Does Not Repair All Axes

10 matched WOD-E2E clips completed by all four variants.

Model	Collision	Offroad	Wrong lane	Progress	Dist. m	Dist.- GT
Raw DAgger iter2	0.600	0.900	0.700	0.034	61.98	13.86
Clamped lateral	0.700	0.500	0.200	0.384	59.85	5.21
Hard veto hybrid	0.800	0.200	0.700	0.837	166.06	22.71
Source decay	0.600	0.900	0.400	0.175	62.92	27.16

What the AlpaSim Interventions Show

Intervention	Repairs	Worsens or fails
Clamped lateral	offroad, wrong-lane, progress, distance-to-GT	collision increases
Hard veto hybrid	progress, offroad	collision increases, wrong-lane unchanged
Source decay	wrong-lane, progress	offroad unchanged, distance-to-GT worsens

The hard veto is especially diagnostic:

- vetoed 1987 / 1990 DAgger argmax decisions
- selected mostly the geometric scorer path

• recovered progress by collapsing toward Spotlight-style selection

Controlled Proxy-State Perturbation

Internal simulator, true-state physics and evaluation fixed.

Only the planner-visible proxy state is corrupted.

Perturbation	Raw p/c/l	Clamp	Hybrid	
clean	0.917/0.083/0.312	0.938/0.062/0.167	0.958/0.042/0.167	0.938,
heading bias	0.896/0.104/0.562	0.938/0.062/0.396	0.938/0.062/0.375	0.896,
actor latency	0.604/0.396/0.875	0.604/0.396/0.521	0.854/0.146/0.208	0.833,
route offset	0.938/0.062/0.229	0.875/0.125/0.021	0.896/0.104/0.062	0.938,

Source: `artifacts/internal_proxy_transfer_medium.md`.

Why the Proxy Sweep Matters

AlpaSim alone leaves an adapter-confound objection.

The proxy sweep removes that escape hatch:

- same simulator dynamics
- same token library
- same controller
- matched scenarios
- only proxy state is corrupted

The same multi-axis tradeoff appears internally under controlled corruption.

That makes the external result mechanistic rather than anecdotal.

The Proposition

Let each candidate token have a metric vector:

```
m(a) = [collision risk, offroad risk, lane risk, progress loss, tracking error]
```

If a proxy-state perturbation changes the action ordering for one metric but not another,

a scalar intervention can improve one axis while worsening another.

The empirical question is not whether this is possible.

The empirical question is whether it occurs repeatedly under realistic AV transfer interfaces.

Our answer: yes, in AlpaSim and in controlled proxy perturbations.

WOD Grounding Probe

479 WOD-E2E validation frames, segment-grouped CV.

Ablation	RFS	Oracle	Regret	Oracle match	Grounding signal
Scalar / geometry only	7.695	9.068	1.373	0.403	none
InternVLA only	7.672	9.225	1.554	0.392	InternVLA
Cosmos + InternVLA linear fusion	7.728	9.208	1.480	0.403	Cosmos + InternVLA
Cosmos 64d nonlinear head	7.845	9.264	1.419	0.390	Cosmos + nonlinear head

Grounding helps only with the right head

Evidence Strength Audit

Current repo audit conclusion: `strong_diagnostic` .

Gate	Current	Target	Status
Internal proxy diagnostic	288 cases	≥ 288 cases	pass
External transfer diagnostic	10 matched scenes	≥ 10 scenes	pass
Larger external scale	10 scenes	≥ 30 scenes	not yet
Uniform positive method	0 external dominance cases	≥ 1	not yet
Grounding prior signal	± 0.150 BFS	≥ 0	pass

Media and Figures Included

Asset	Use
<code>docs/images/baseline_spotlight.gif</code>	baseline visual failure
<code>docs/images/spotlight_success.gif</code>	geometry-grounded success
<code>docs/images/construction_success.gif</code>	long-tail construction case
<code>docs/images/intersection_stress.gif</code>	multi-actor stress case
<code>docs/images/alpasim_reasoning_panel.png</code>	AlpaSim adapter reasoning
<code>docs/images/dagger_aggregation_ablation.svg</code>	Dagger aggregation plot
<code>docs/images/alpasim-bridge.svg</code>	external bridge diagram

GIFs are used as short videos in GitHub, Marp, and most slide renderers.

Additional Visual Cases



These examples are not the proof.

They show what the controlled simulator is stressing: narrow corridors, route

Reproducibility Path for Auditors

Start with the audit wrapper:

```
./scripts/run_corl2027_audit.sh
```

Core evidence files:

```
docs/corl2027/paper.tex  
docs/corl2027/AUDIT.md  
artifacts/corl_evidence_strength_audit.md  
artifacts/internal_proxy_transfer_medium.md  
artifacts/alpasim_matrix10_analysis.md  
artifacts/wod_grounding_ablation_table.md
```

AlpaSim setup and runtime are documented in `docs/notes/alpasim-integration.md`.

Submission Framing

Do not frame this as:

“Our hybrid method dominates.”

The data does not support that.

Frame it as:

“Internal imitation-learning success can hide transfer-axis disagreement.

We provide a controlled diagnostic, reproduce the effect externally in AlpaSim, and show that physical grounding must be evaluated per axis rather than as one aggregate planning score.”

That is the defensible CoRL contribution.

Takeaway

The surprising result is not that DAgger fails externally.

The result is where it fails:

- the internal simulator says the learned token policy is calibrated
- AlpaSim shows the selected tokens are often externally incompatible
- controlled proxy corruption reproduces the same axis conflicts
- frozen world/VLA priors help only through head-sensitive grounded selection

Robust AV evaluation needs per-axis transfer diagnostics, not only offline accuracy or aggregate closed-loop pass rate.